

The Feynman Lectures on Physics, Volume II

Problem Set 5

Due 2022.5.15

1 Solutions of Maxwell's Equations in Free Space (6pt)

It is sometimes convenient to consider complex solutions of differential equations.

- (a) If we assume the fields vary sinusoidally in time and with the coordinate x (no y - nor z -dependence), show that each component of, for example

$$\mathbf{E} = \mathbf{E}_0 e^{+i(\omega t - kx)}$$

satisfies the wave equation. (Remember that the actual field is found by taking the real part of this expression.)

- (b) Convince yourself that the real part of $\mathbf{E}$ corresponds to a plane wave traveling along the x -axis. In what direction is it going?
- (c) Show that the operation ∇ when applied to a function like that in part (a) becomes

$$\nabla = \mathbf{e}_x \frac{\partial}{\partial x} = \mathbf{e}_x (-ik)$$

where $\mathbf{e}_x$ is a unit vector along the x -axis and i is $\sqrt{-1}$; i.e. show that we can replace the ∇ operation by a simple multiplication.

- (d) What similar statement can you make about the time derivative applied to a function like that in part (a)?
- (e) Using the results of part (c) above write down (by inspection) how Maxwell's equations appear when applied to fields which vary sinusoidally with x and t . What relationship must exist between k and ω ?
- (f) If the field has the form $\mathbf{E} = \mathbf{E}_0 e^{+i(\omega t + kx)}$ how do your answers change?

2 Solutions of Maxwell's Equations with Currents and Charges (5pt)

A particle with charge q moves in a circle of radius a with a uniform speed v , as shown in Fig 1. Consider the fields when the particle is at point P :

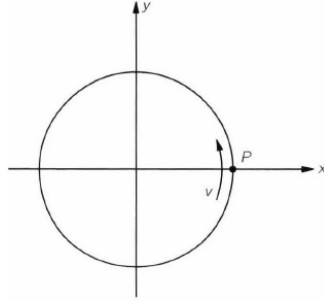


Figure 1:

- Find the potential ϕ at the center of the circle.
- Find the vector potential $\mathbf{A}$ at the center.
- Determine the electric field $\mathbf{E}$ and magnetic field $\mathbf{B}$ at the center using the potentials and Eqs. (18.19) and (18.16) in Vol. II.
- What is the direction of the electric field with respect to the radius vector to point P ?
- Also calculate these fields using Eq. (21.1) in Vol. II.

Note: this problem is relativistic. The speed v is not necessarily small compared to c .

3 AC Circuits (4pt)

- Find the current I in the circuit shown in Fig 2.
- What is the current I if there is a mutual inductance $\mathcal{M}$ which couples the two inductances?

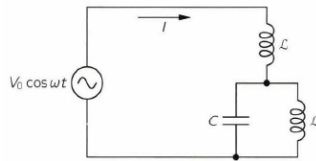


Figure 2:

4 Waveguides (6pt)

The electric fields inside of waveguides which are described in Chapter 24 of Vol. II have the property that the component of the electric field in the direction of propagation is zero; i.e. , the electric field is transverse. (Modes of propagation such as these are therefore called TE, or transverse electric, modes.) There are also modes called TM modes in which there is no magnetic field in the direction of propagation. For the rectangular wave guide shown in Figs. 24-3 and 24-4 in Vol. II, the vector potential of the TM modes is given by

$$\mathbf{A} = \mathbf{e}_z \sin \frac{m\pi x}{a} \sin \frac{n\pi y}{b} e^{i(\omega t - k_z z)}$$

where

$$k_z = \sqrt{\left(\frac{\omega}{c}\right)^2 - \left(\frac{m\pi}{a}\right)^2 - \left(\frac{n\pi}{b}\right)^2}$$

- (a) Satisfy yourself that the magnetic fields found from this are really transverse.
- (b) Show that the $\mathbf{E}$ and $\mathbf{B}$ fields satisfy the wave equation and the proper boundary conditions.
Hint: We require that

$$\begin{aligned}\mathbf{E} &= -\nabla\phi - \frac{\partial\mathbf{A}}{\partial t} \\ \mathbf{B} &= \nabla \times \mathbf{A}\end{aligned}$$

where

$$\nabla \cdot \mathbf{A} = -\frac{1}{c^2} \frac{\partial\phi}{\partial t}$$

- (c) Show that the mn -mode is not propagated if

$$\omega < c\sqrt{\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2}$$

5 Electrodynamics in Relativistic Notation (2pt)

A positron (e^+) can be made by bombarding a stationary electron (e^-) with a photon (γ),

$$\gamma + e^- \rightarrow e^- + e^+ + e^-$$

What is the minimum photon energy E^γ . Use four-vectors and invariant combinations of them wherever possible.

6 Lorentz Transformations of the Fields (4pt)

A very long straight wire carries a current I produced by electrons moving at velocity $\mathbf{v}$. Stationary positive ions in the wire make the total charge density vanish.

- (a) Find the fields outside the wire in a frame stationary with respect to the wire.
- (b) Transform the fields to a frame moving with the electrons.
Hint: In Chapter 13 of Vol. II the electric field observed from this moving frame was obtained by another method; see Eq. (13.28).